

PLABABLE

GEMS 

VERSION 1.3

EMERGENCY MEDICINE



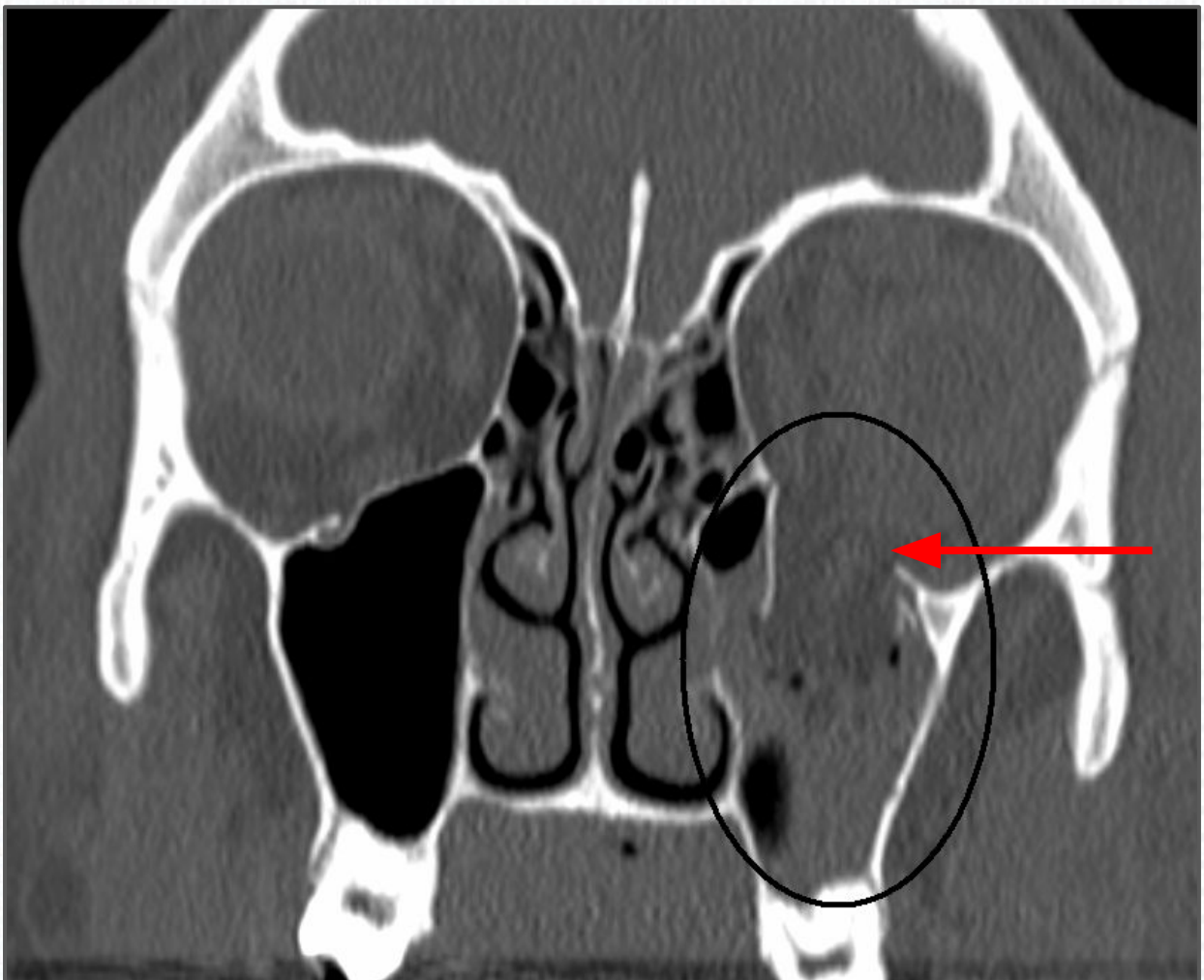
Orbital Blowout Fracture

Presentation

- Blow from an object → **orbital floor (maxilla) #**
- **Vertical diplopia** especially on looking up (impingement of the inferior rectus muscle)
- Enophthalmos (sunken eye)
- Infraorbital anaesthesia

Management

- X-ray of the skull
- CT scan (gold standard)
- Surgical management



Head Injury (adults)

In the United Kingdom there are stringent requirements for performing a head CT scan

These requirements are set according to NICE guidelines

CT scan indicated within 1 hour

- GCS < 13 on initial assessment in ED
- GCS < 15 at 2 hours after the injury on assessment in the ED
- Suspected open or depressed skull fracture
- Any sign of basal skull fracture
- More than 1 vomiting episode
- Post-traumatic seizure
- Focal neurological deficit

CT scan indicated within 8 hours

- 65 years or older
- Any history of bleeding or clotting disorders
- Dangerous mechanism of injury

Remember:
if GCS is less than 8 you must intubate

Head Injury (children)

In the United Kingdom there are stringent requirements for performing a head CT scan
These requirements are set according to NICE guidelines

CT scan indicated within 1 hour if any of the following applies

- Post-traumatic seizure but no history of epilepsy
- On initial assessment in the ED, GCS < 14 (less than 15 if the child is under 1 year old)
- GCS < 15 at 2 hours after injury
- Suspected open or depressed skull fracture or tense fontanelle
- Any sign of basal skull fracture
- Focal neurological deficit

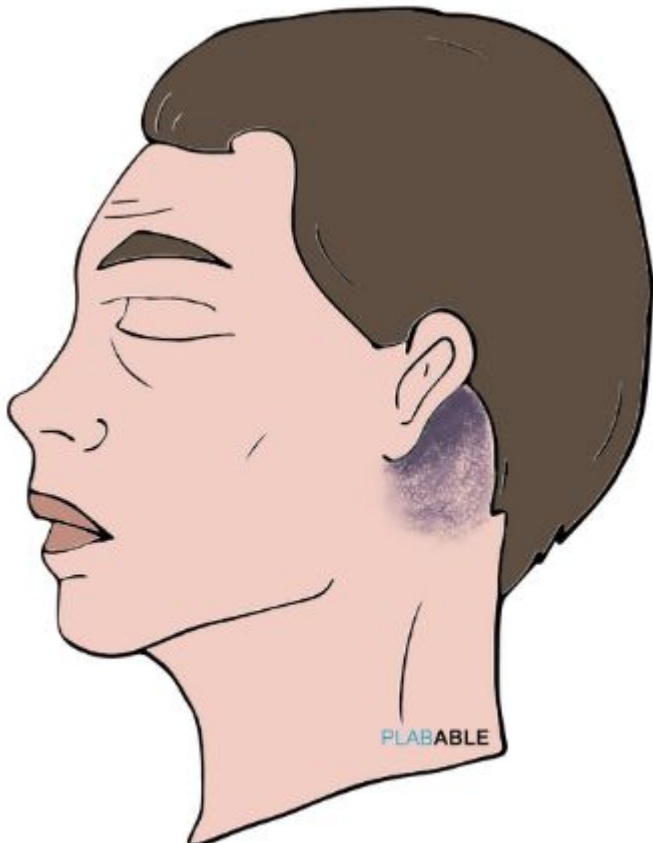
CT scan indicated within 1 hour if 2 of the following applies

- Loss of consciousness lasting more than 5 minutes
- Abnormal drowsiness
- Three or more discrete episodes of vomiting
- High-speed road traffic accident
- Fall from a height of greater than 3 metres

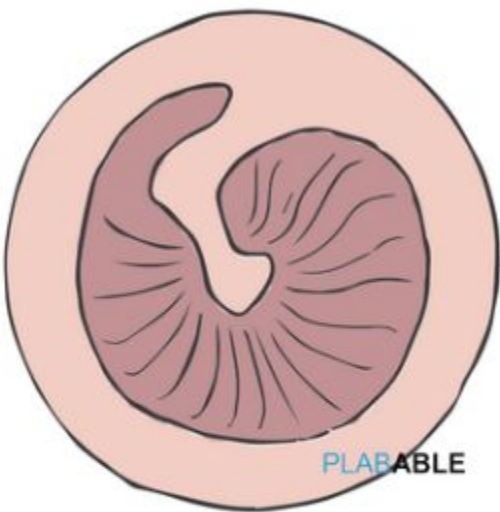
Basal skull fracture



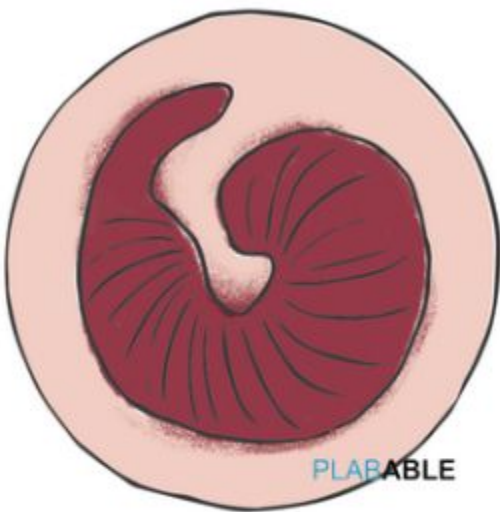
Raccoon Eyes & CSF Rhinorrhoea



Battle Sign



Typical Membrane



Haemotympanum

Head Injury

Brain trainer:

A factory worker has hit his head on some machinery and but not lost consciousness. He is not able to recall the mechanism of injury nor has he vomited. When is head CT scan indicated?

➔ **Within 8 hours**

Due to the presence of retrograde amnesia and a dangerous mechanism of injury

Pupillary Response

Brain trainer:

A patient has a unilaterally dilated pupil. What diagnosis might you expect?

➔ **Space occupying lesion
(e.g. abscess, tumour, haematoma)**

Pupillary Response

Brain trainer:

A patient has bilateral mydriasis (dilated pupils).
What diagnoses are high yield for the exam?

→ **Stimulants / tricyclic antidepressant overdose**

Pupillary Response

Brain trainer:

A patient has bilateral miosis (constricted pupils). What diagnoses are high yield for the exam?

→ Opiate overdose, CVA of the brainstem

Drug Overdoses

Drug	Presentation	Treatment
Paracetamol	RUQ pain Vomiting Jaundice Liver failure	<1 hr: activated charcoal 4hr plasma value > treatment line OR staggered dosing: IV N-acetylcysteine
Aspirin	Vomiting Hyperventilation Tinnitus Vertigo Respiratory alkalosis → Metabolic acidosis	<1 hr: activated charcoal Metabolic acidosis: alkalinization of urine Salicylate level >700: dialysis
Opioids	Miosis Respiratory depression Low BP and HR	IV naloxone

Heroin

Brain trainer:

A man is brought into the ED by a group of his friends. He had been at a party and ingested some drugs. He is hypotensive, with RR 6 and constricted pupils. What is the most likely diagnosis?

➔ **Heroin overdose**

Aspirin

Brain trainer:

Which commonly prescribed over the counter medication initially causes a respiratory alkalosis and then later a metabolic acidosis?

→ Salicylate (aspirin) poisoning

Drug Overdoses

Drug	Presentation	Treatment
Organo-phosphate	SLUD Salivation Lacrimation Urination Diarrhoea	IV Atropine IV Pralidoxime
TCA - Amitriptyline	Widened QRS and broad complex tachycardia	Sodium bicarbonate 0.9% NS (hypotension)
Benzodia-zepines	Respiratory depression	Flumazenil
Cocaine	High BP, HR and RR Mydriasis Perforated nasal septum MI	Benzodiazepines
Cyanide	Dizziness Anxiety Tachycardia Vomiting	100% oxygen Sodium nitrite/ sodium thiosulfate Hydroxocobalamin

King's College Hospital Criteria

**With a paracetamol overdose we must arrange
for immediate liver transplantation
if the criterias are met**

Criteria

Arterial pH < 7.3, 24 hours after ingestion or all of the following:

- Prothrombin time > 100 seconds
- Creatinine > 300 $\mu\text{mol/l}$
- Grade III or IV encephalopathy

Benzodiazepines

Brain trainer:

An overdose of benzodiazepines will cause what acid-base disturbance?

→ **Respiratory acidosis (RA)**

(Apnea → Accumulation of CO₂ → RA)

Stages of Hypovolaemia

For adults, the clinical staging relating to loss of blood volume can be classified as:

	Stage 1	Stage 2	Stage 3	Stage 4
Blood loss	10-15%	15-30%	30-40%	Over 40%
Blood pressure	Normal	Postural hypotension	Hypotension	Marked hypotension
Heart rate	Normal	Slight tachycardia (> 100 bpm)	Tachycardia (> 120 bpm)	Extreme tachycardia (>140 bpm)
Respiratory rate	Normal	Increased (> 20)	Tachypnoea (> 30)	Extreme tachypnoea
Mental status	Normal	Slight anxiety, restless	Altered, confused	Decreased consciousness, lethargy, or coma
Urine output	Normal	20-30 mL/hour	Less than 20 ml/hour	No urine output

Plabable’s Tip
Remember the heart rate and corresponding stage

Stages of Hypovolaemia

Brain trainer:

A butcher has cut his leg and is bleeding profusely. His heart rate is 130. What percentage of blood has he lost?

→ 30-40%

Opioid Overdose

Presentation

- Pinpoint pupils
- Respiratory depression
- Coma

Management

- IV Naloxone



Opioid withdrawal

- Sweating, rhinorrhoea, tremor, dilated pupils, tachycardia and hypertension
- **Detoxification:** methadone
- **Relapse prevention:** naltrexone

Drug Withdrawal

Drug	Withdrawal symptoms	Treatment
Opioids	- Sweating - Rhinorrhoea - Tremor - Dilated pupils - Tachycardia - Hypertension	Methadone
Benzo-diazepines	- Anxiety - Insomnia - Agitation	Diazepam and slowly reduce the dose Propranolol
Cocaine	- Depression - Restlessness	Propranolol Diazepam
Alcohol	- Insomnia and fatigue - Tremor - Anxiety/feeling nervous - Nausea and vomiting - Palpitations - Alcohol hallucinosis - Seizures	Chlordiazepoxide IV thiamine

Carbon Monoxide (CO) Poisoning

- CO is produced by incomplete combustion (car exhausts and burning home)
- It reduces the oxygen carrying capacity by binding to Hb (CO has more affinity for Hb than O₂)

Presentation

- Headache
- Nausea and vomiting
- Vertigo

Investigation

- HbCO level in blood

Management

- 100% oxygen
- Use of hyperbaric oxygen is controversial



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Superficial Burn



Red and painful but do not blister

Partial Thickness Burn



**Range from blistering to deep dermal burn.
The appearance is shiny and sensation is intact.
Capillary refill blanches.**

Full Thickness Burn



Has a leathery or waxy appearance. It may be white, brown or black in colour. There are no blisters and sensation is lost so they do not feel any pain. There is no capillary refill.

Burns

Brain trainer:

Above what percentage of total body surface area burned requires fluid replacement?

→ Children - 10%

→ Adults - 15%

Smoke Inhalation Injury

Presentation

- Persistent cough
- Stridor
- Wheezing
- Black sputum suggests excessive exposure to soot
- Use of accessory muscles of respiration
- Blistering or oedema of the oropharynx
- Hypoxia or hypercapnia

Management

- Oxygen
- Intubation
- IV fluids

Parkland Formula

Brain trainer:

What is the formula used for calculating fluids for burn patients?

→ 4 x weight (kgs) x percentage of area of burn (ml of fluids)

Hypoglycaemia

Etiology

- Hypoglycaemics, binging of alcohol, post gastric surgery

Presentation

- Autonomic: sweating, anxiety, hunger, tremor, palpitations, dizziness
- Neuroglycopenic: confusion, drowsiness, visual trouble, seizures, coma

Management

- If conscious + orientated
→ carbohydrate snack
- If conscious + uncooperative
→ glucose gel between gums
- If unconscious
→ glucagon 1mg IV/IM or glucose IV

Alcoholic

Brain trainer:

What laboratory findings would you expect in a chronic alcoholic who is vomiting profusely?

→ Hypoglycemia

→ Hypokalemia

Mallory-Weiss Syndrome

Presentation / Etiology

1. Persistent vomiting/retching
 2. Oesophageal tear
 3. Haematemesis
- Alcoholism
 - Bulimia nervosa

Investigation

- Endoscopy

Management

- Maintain airway, provide high-flow oxygen, correct fluid losses by giving IV fluids. Intravenous blood can also be given in severe cases.
- Endoscopy immediately after resuscitation to confirm diagnosis and stop any potential further bleeding.

Oesophageal Varices

Management

- Always start with fluid resuscitation
- Endoscopy should be undertaken immediately after resuscitation for unstable patients with severe acute upper gastrointestinal bleeding. This allows the bleeding point to be visualised and treated
- Band ligation is the first choice treatment for all patients with upper gastrointestinal bleeding from oesophageal varices
- Administer terlipressin at presentation (2mg intravenously repeated every 4 to 6 hours). Treatment should be stopped after definitive haemostasis has been achieved or after five days
- Avoid omeprazole acutely unless the patient has known peptic ulcer disease. PPIs should not be used prior to diagnosis by endoscopy

Choking Management

Infant

1. Whilst seated hold infant in prone position with head down (gravity aids in removing foreign body) deliver 5 back blows with heel of hand along spinal cord (between shoulder blades).
2. Turn infant into a supine position and deliver five chest thrusts. These thrusts are similar to CPR but sharper in nature and lower in frequency.

Child / adult

1. Procedure same as for infants (i.e. back blows)
2. Give 5 abdominal thrusts - stand behind victim who is leaning forward, put both hands around upper abdomen and clench one fist and grip this fist with other hand. Pull upwards and inwards sharply.
3. Alternate between step 1 and step 2

If unconscious

1. Place victim on floor
2. Call ambulance
3. Begin CPR

Acute Alcohol Withdrawal

Presentation

- **Insomnia and fatigue**
- **Tremor**
- **Anxiety/feeling nervous**
- **Nausea and vomiting**
- **Excessive sweating**
- **Palpitations**
- **Alcohol hallucinosis**
- **Seizures**

Management

- **Benzodiazepines (chlordiazepoxide)**
- **IV thiamine**



Delirium Tremens

Delirium tremens

Usually seen in alcoholics 24-72 hrs after hospital admission because of withdrawal

- **Hallucinations**
- Confusion
- **Delusion**
- Severe agitation
- Seizures

Management

- Benzodiazepines (**diazepam**)
- IV thiamine



Wernicke's Encephalopathy

Disorder due to **thiamine** (Vit B1) deficiency
mainly in chronic alcoholics

Ophthalmoplegia
(Double vision)



Ataxia

Confusion

Treatment

- IV thiamine followed by glucose



Common in
chronic alcoholic



Thiamine (B1)
deficiency

Korsakoff syndrome

- Anterograde amnesia
- Confabulation (making up stories)

Acute Compartment Syndrome

Trauma causing $\uparrow$ pressure within an anatomical space compromising the blood flow and causing ischemia

Presentation

- Increasing pain despite analgesia
- Sensory deficit
- Excessive pain on passive movement
- If severe: Pallor, pulselessness, paralysis, coolness and loss of capillary return

Investigation

- Intracompartmental pressure measurement

Treatment

- Emergency **fasciotomy**



Fasciotomy

Acute Compartment Syndrome

Reperfusion injury

Once the ischemic tissue is perfused after fasciotomy, myoglobin is released into circulation
→ **acute kidney injury**

Prevention

- Adequate hydration and alkalinisation of urine
- In severe cases dialysis is required

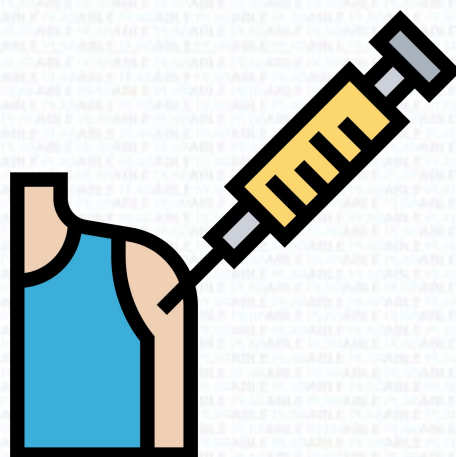
Anaphylaxis

Presentation

- **Respiratory:**
 - Exposure to allergen (peanuts, eggs etc.)
 - Swelling of lips, tongue, pharynx and epiglottis potentially result in complete upper respiratory tract obstruction
 - Wheeze
 - Hypoxia and hypercapnia
- **Skin:** Pruritus, urticaria and angioedema
- **CVS:** Hypotension and shock

Management

- **IM adrenaline**
 - Adults: 0.5ml of 1 in 1000
 - Children up to 6 yrs: 0.15 ml of 1 in 1000
 - Children 6 - 12 yrs: 0.3 ml of 1 in 1000



IM adrenaline

Panic Attack

Presentation

- Periods of intense fear usually for 10-20 min associated with:
 - Palpitations
 - Sweating
 - Tremors
 - Difficulty breathing
 - Chest pain
 - Numbness or tingling sensation
(Hyperventilation → CO₂ washout → Respiratory alkalosis → Hypocalcaemia)

Diagnosis requires attack followed by ≥ 1 month of ≥ 1 of the following:

- Persistent concern of additional attacks
- Worrying about consequences of attack
- Behavioral change related to attacks

Management

- Acute: Breathing through the bag
- Long term: CBT, antidepressants (SSRI)

Flail Chest

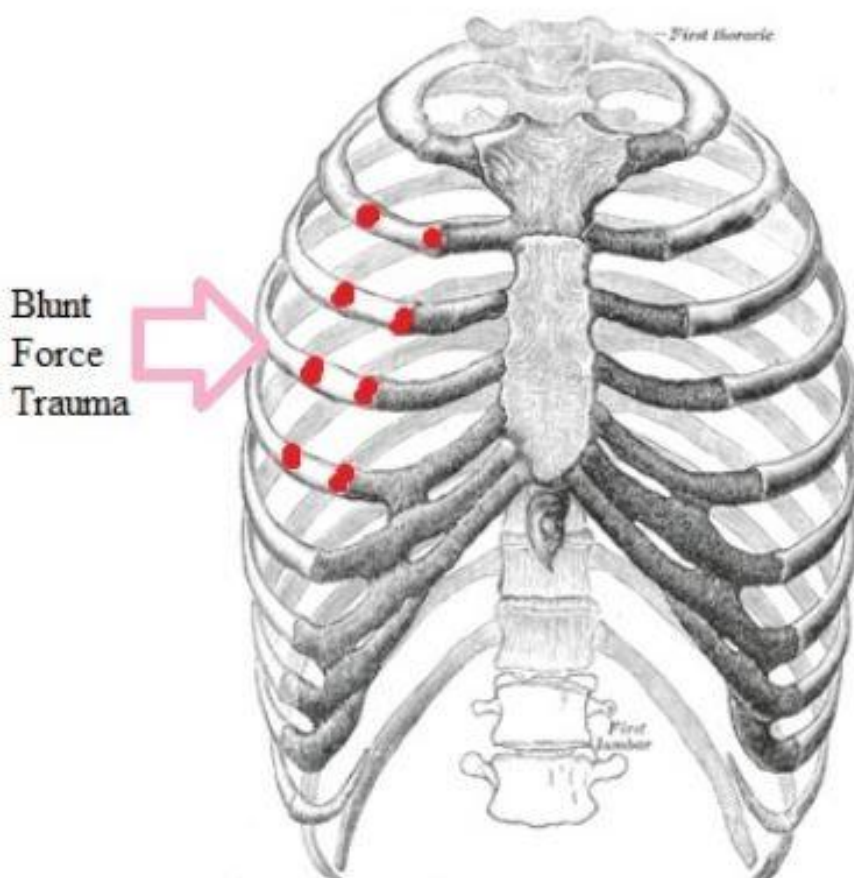
Multiple broken ribs due to trauma which move in during inspiration and out during expiration causing **paradoxical breathing**

Presentation

- RTA
- Chest pain
- Shortness of breath
- Associated with pulmonary contusion and pneumothorax

Management

- Adequate analgesia
- Positive pressure ventilation (If severe)



Hereditary Angioedema

Autosomal dominant condition due to
C1 esterase deficiency

Presentation

- Facial, lip and mouth swelling
- Laryngeal oedema - life threatening (dyspnoea and stridor)
- Abdominal pain
- Precipitated by stress, infection, trauma/surgery

Investigation

- Serum C4 level
- C1 Inhibitor level

Management

- **IV or SC C1 inhibitor concentrate** (first line for acute treatment and prophylaxis)
- Bradykinin receptor inhibitor
- Kallikrein inhibitor
- Severe laryngeal oedema requires intubation and ventilation

Note: Anaphylaxis have a similar presentation and is treated with **IM adrenaline**. Allergy and family history is important to differentiate both.

Perforated Peptic Ulcer

Presentation

- Epigastric pain
- Haematemesis
- Malena

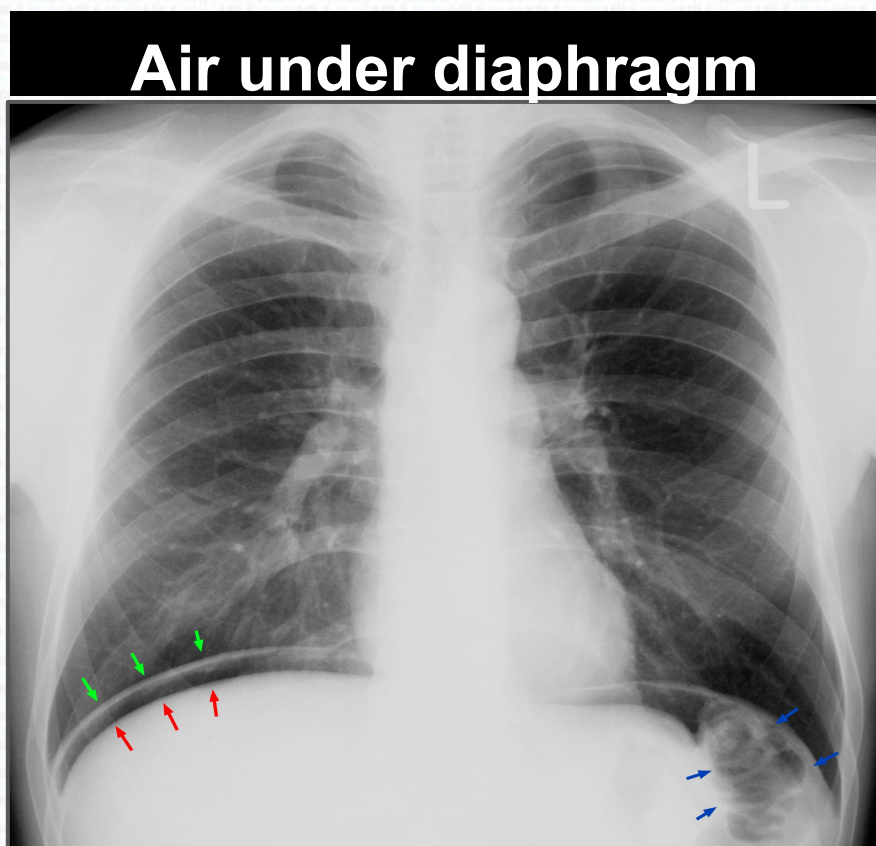
Risk factors: NSAIDs, steroids, *H. pylori* infection

Investigation

- **Erect X-ray abdomen:** air under diaphragm
- Endoscopy

Treatment

- Endoscopic repair



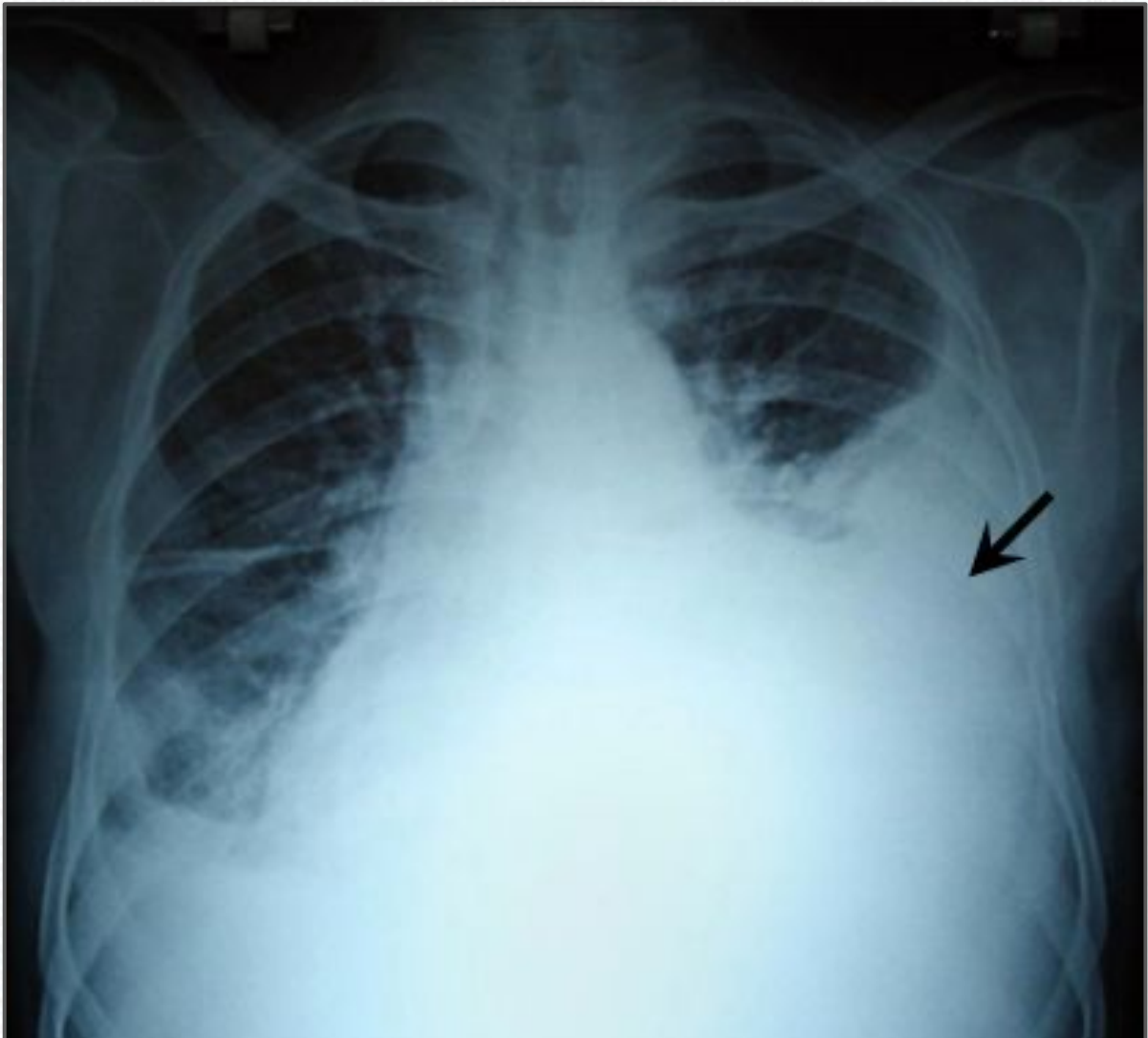
Haemothorax

Presentation:

- RTA or stab injury
- Hypotension and tachycardia
- Dullness on percussion
- Chest X-ray: homogenous opacity on the lower region of the lung

Management

- Chest drain
- Surgical exploration



Pneumothorax

Collection of air in the pleural cavity resulting in collapse of the lung on the affected side

Causes

- Primary in healthy individuals (smoking and **marfan** syndrome are risk factors)
- Secondary in RTA, stab injuries and COPD

Presentation

- Sudden onset of chest pain
- Shortness of breath
- Hyperresonance in the affected side
- **Tension pneumothorax:** Pulsus paradoxus, tracheal deviation and hypotension

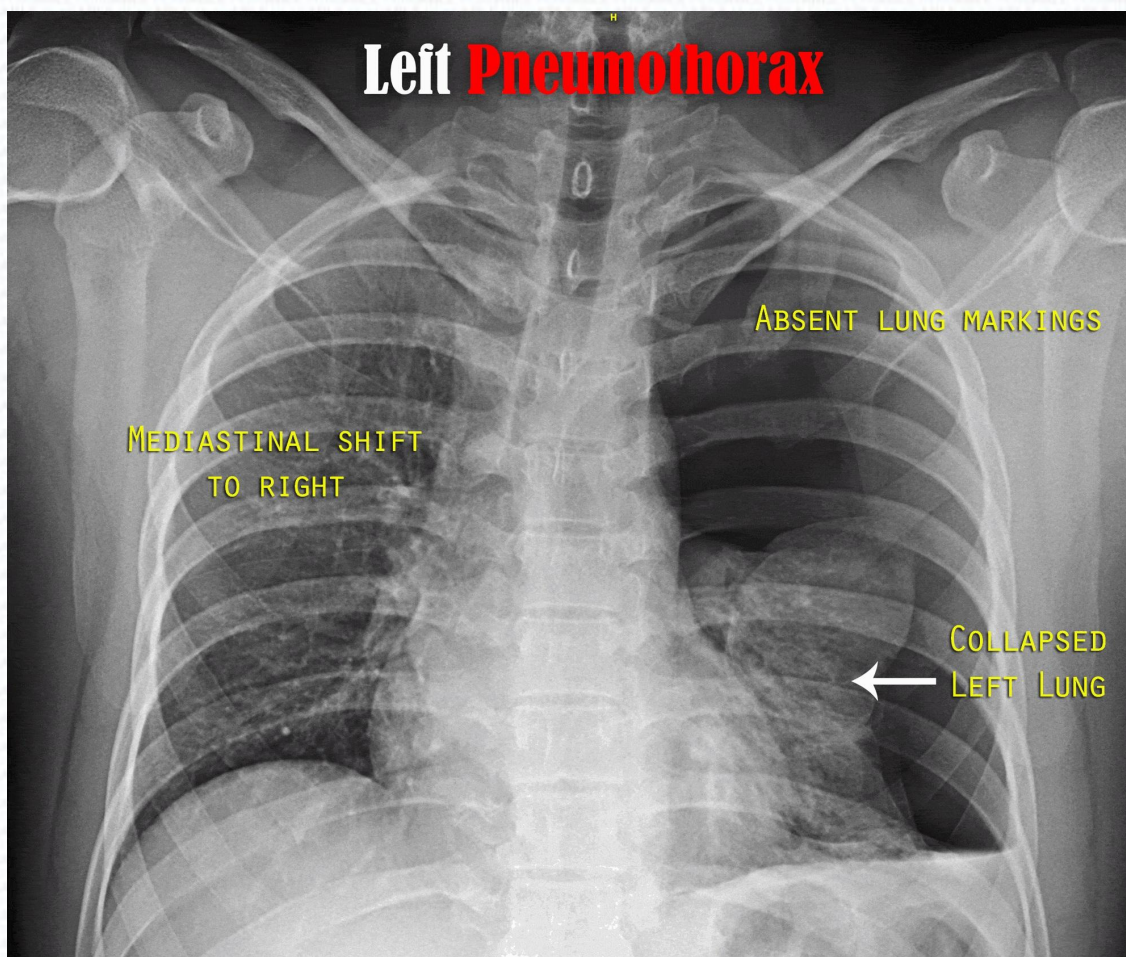
Pneumothorax

X-ray:

Hyperlucent lung, tracheal deviation and collapse of lung

Management:

- **Tension pneumothorax:** needle decompression
- **Mild cases:** supplemental oxygen and observation



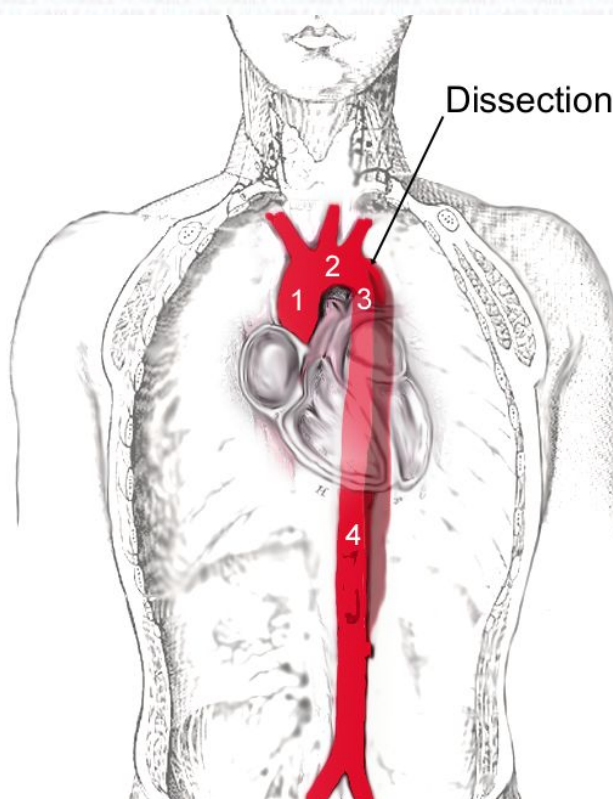
Aortic Dissection

Presentation

- Severe chest pain radiating to the back
- Loss of pulse
- Rupture:
 - Hypotension
 - Pericardial cavity - cardiac tamponade
 - Pleural cavity - haemothorax
 - Mediastinum

Risk factors

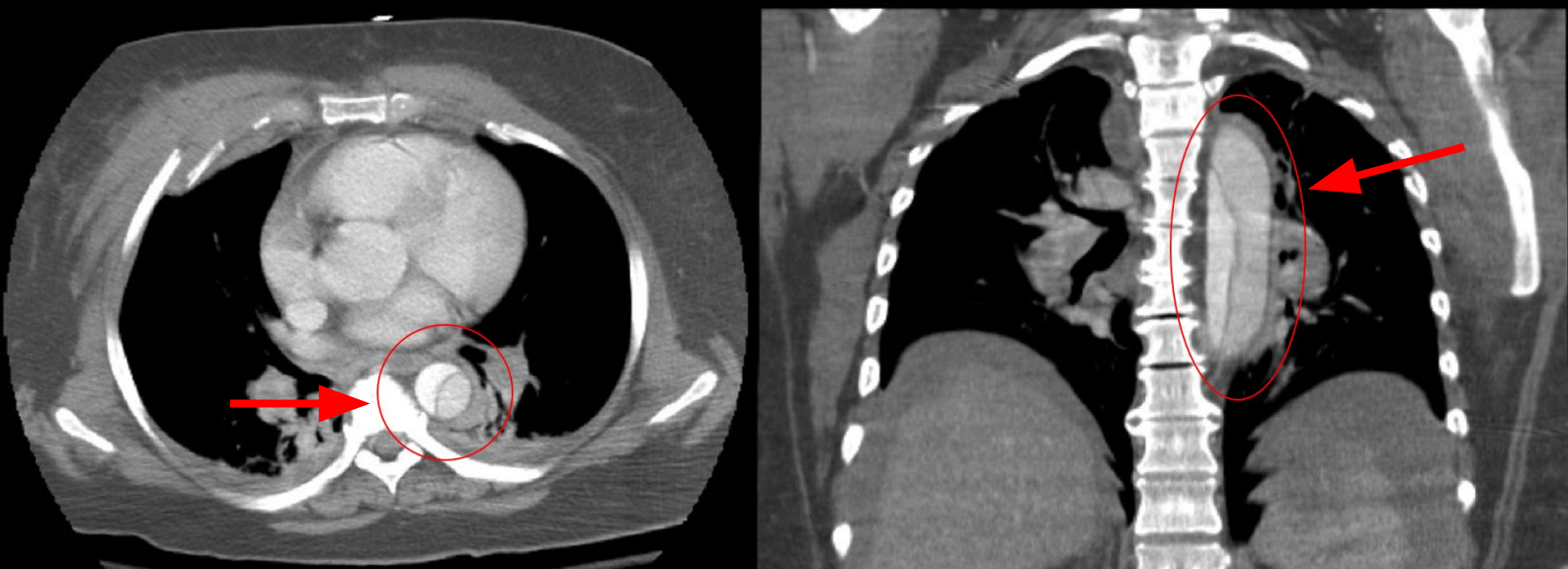
- Hypertension
- Smoking
- Marfan's syndrome
- Ehlers-Danlos syndrome



Aortic Dissection

Investigations

- Transoesophageal USG (immediate)
- MRI (gold standard)



Descending Aortic dissection

Management

- Control hypertension with beta blockers
- Surgery - grafting or stenting

Note: When the patient is having unstable vitals in any clinical scenario always think of ABC as the first choice

Sepsis

Red Flag for sepsis based on NICE:

- **Systolic BP** <90 mm Hg
(or >40 mm Hg fall from baseline)
- **HR** >130 beats per minute
- **SpO2** <91%
- **RR** >25/min
- Responds only to voice or pain/unresponsive
- **Lactate** >2.0 mmol

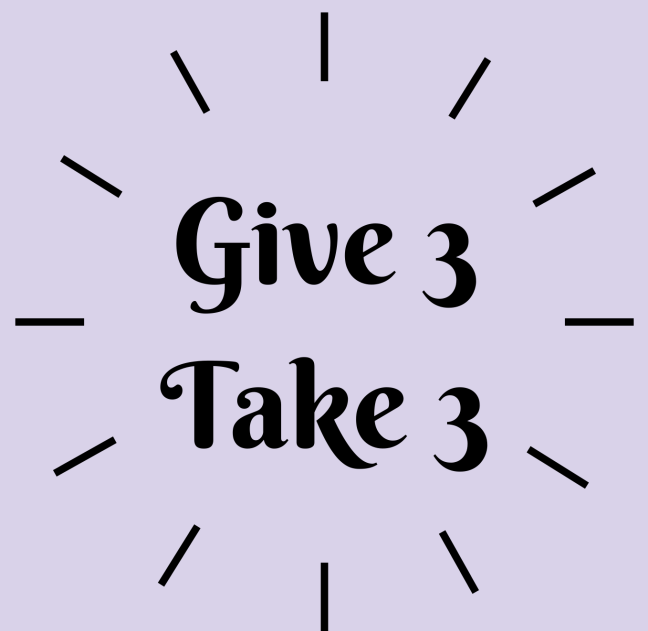
Sepsis six to be started if any red flag is present:

Give 3:

- **Fluid** resuscitation
- High-flow **oxygen**
- **IV antibiotics**

Take 3:

- **Blood culture**
- Hb and serial **lactates**
- Hourly **urine output**



Toxic Shock Syndrome

Multisystem inflammatory response to bacterial exotoxins (*staphylococci* and *streptococci*)

Presentation

- Fever
- Hypotension
- Diffuse erythematous rash
- Multiorgan dysfunction
- Desquamation of palms and soles after 1-2 weeks
- Risk factor: **tampon** usage

Treatment is same as sepsis along with steroids

Serotonin syndrome

↑ levels of serotonin in the synapses due to drug overdose or interaction between **MAOI, SSRI or SNRI**

Presentation:

- **Autonomic hyperactivity**
 - Hypertension
 - Hyperthermia
 - Tachycardia
- **Neuromuscular abnormality**
 - Tremor
 - Ocular clonus
 - Hypertonicity
- **Mental status changes**
 - Anxiety
 - Confusion

Management:

- Withdrawing offending drug
- IV fluids and benzodiazepines
- Cyproheptadine

Neuroleptic Malignant Syndrome

Life threatening condition due to potent neuroleptics such as haloperidol and fluphenazine

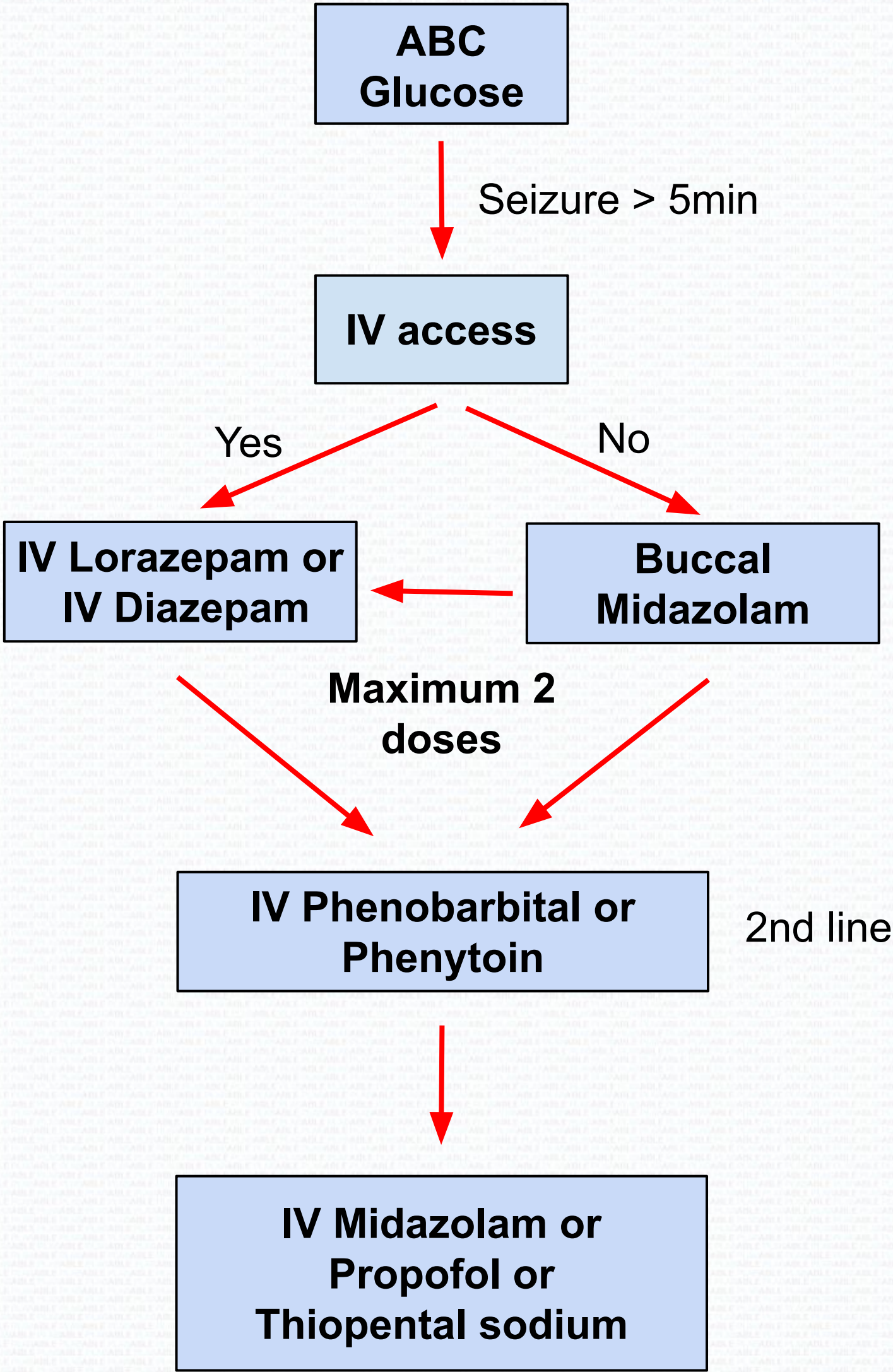
Presentation

- Hyperthermia
- Muscular rigidity
- Autonomic instability

Management

- Stop the drug
- Benzodiazepines
- Dantrolene sodium (muscle relaxant)

Generalised Tonic Clonic Seizure



Resuscitation - Adult

30 Chest compression



2 Rescue breaths



**Continue CPR
30:2**

Resuscitation - Child

Open airway



5 Rescue breaths



15 Chest compression



**2 Rescue breaths
15 Chest compression**

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